

Analysis of sociodemographic and linguistic influences on verbal fluency performance in Basque and Catalan bilingual adults

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The aim of the study was to generate norms for letter, semantic, and verb fluency tests for Basque and Catalan speakers from Spain. The sample consisted of 275 bilinguals aged between 18 and 85 years who completed the tests for each language (Basque–Spanish or Catalan–Spanish) with order counterbalanced. The influence of predictors was evaluated with Generalized Linear Mixed Models. Language by letter and type of language by semantic category interactions were found, where people scored higher in letter fluency when tested in Spanish than in Basque or Catalan, and produced more words in the “animals” category compared to “fruits” and “occupations”. Production increased in all tests with higher education. Letter and semantic fluency scores increased until ages 40–45, after which they decreased, except for action fluency, where Basque speakers had decreased scores with age. Basque men scored higher than Basque women in letter and semantic fluency, but lower in verb fluency; while Catalan women scored higher than Catalan men in semantic and verb fluency, but lower in letter fluency. The study provides, for the first time, verbal fluency norms for Basque and enriches norms availability for Catalan. Socio-psycholinguistics aspects may have an impact on performance.

Introduction

Verbal fluency (VF) tests are commonly used in the clinical work and research as part of neuropsychological batteries (i.e. the Consortium to Establish a Registry for Alzheimer’s Disease,

the Addenbrooke's Cognitive Examination) because of its ease and rapid administration, and its sensitivity to brain damage (Gollan, Montoya and Werner 2002; Azuma 2004). Therefore, it has been used to detect neurological damage in multiple disorders, such as Alzheimer's (Teng et al. 2013), Parkinson's (Herrera, Cuertos and Ribacoba 2012), Vascular (Macoir 2023), or Huntington's diseases (De Lucia et al. 2020), mild cognitive impairment (Teng et al. 2013), traumatic brain injury (Zakzanis, McDonald and Troyer 2013), stroke (Heshmatollah et al. 2021), among others.

A variety of VF tests exists (i.e. excluded letter VF, alternating VF), although phonological/letter and semantic/category forms are the most widely used. In both cases, participants require the ability to generate words that meet specific rules in a given time, usually 1 minute. Although both VF forms are similar, the underlying cognitive, and, therefore, anatomical neuronal substrates differ somehow. For semantic/category VF, participants evoke words by the activation of semantic network, so strategic organization and lexical knowledge are needed (Azuma 2004; Federmeier, Kutas and Schul 2010). However, during phonological/letter VF, words must be produced attending to phoneme or letter network, which is not a common strategy for word retrieval. These strategies are less useful and, instead, a higher order functions are required to inhibit automatically semantic associations and to update and shift from responses using searching strategies based on phonemic or letter cues (Brandeker and Thordardottir 2023). Therefore, phonological/letter VF is a more difficult task or effortful than semantic/category fluency. During both tasks, participants must be focused to keep in memory the rules and previous responses to avoid incorrect responses and repetitions, as well as to address the limited time to word production (Reverberi et al. 2014). Regarding brain networks, literature has found the implication of the frontal lobe during the performance of phonological/letter fluency, and the temporal lobe during semantic/category fluency, but it is more likely that VF requires coordinated activation of several brain areas (Wagner et al. 2014).

In addition to the aforementioned tests, there is also the action/verb fluency test, which is less commonly used despite its potential to discriminate not only between healthy individuals and those with dementia, but also among those with mild cognitive impairment that present high probability to convert to dementia (Alegret et al. 2018). This may be because verbs are more complex than nouns at morphological, syntactic, and semantic levels (Vigliocco et al. 2011; Feng et al. 2020), making more effortful its production. Moreover, it seems that verb fluency is particularly sensitive to frontal-subcortical networks (Paek, Murray and Newman 2020).

Different variables have been associated with the performance on these VF tests. Formal education and age are the most important variables; young individuals with higher educational levels tend to produce more words (i.e. Olabarrieta-Landa et al. 2015; Marquine et al. 2021; Jebahi, Abou Jaoude and Ellis 2022; Zegarra-Valdivia, Chino and Paredes Manrique 2022; Macoir and Hudon 2023). There is no consensus regarding sex (i.e. Carvalho and Caramelli 2020; Marquine et al. 2021; Iñesta, Oltra-Cucarella and Sitges-Maciá 2022; Shirdel et al. 2022), but despite the mixed finding, its influence is very small compared to age and education. In addition, the literature also describes the influence of bilingualism, particularly in VF test, because it requires language proficiency and executive control (Friesen et al. 2015). Bilingualism has been associated with better performance in nonverbal executive function test compared to monolingualism (Bialystok 2009), but mixed findings have been reported regarding executive function tests that have a verbal component, such as VF tests. Some studies have described equivalent performances among monolinguals and bilinguals during both tasks (Blumenfeld, Bobb and Marian 2016), others have found equal or higher performance of bilingual participants on phonological/letter task (Portocarrero, Burright and Donovanick 2007; Bialystok, Craik and Luk 2008; Friesen et al. 2015; Brandeker and Thordardottir 2023), but poorer performance in semantic/category fluency tests compared to monolinguals (Portocarrero, Burright and Donovanick 2007), while some studies have reported disadvantages that persist even when participants complete the task in their dominant language (Gollan et al. 2008). The explanation for these difficulties during lexical access, especially during semantic/category fluency, has been linked to smaller vocabulary size (Luo, Luk and Bialystok 2010)

and between-language interference that produces greater selection demands and slower production of words (Sandoval et al. 2010).

During recent years, an increased number of studies around the world have focused on obtaining normative data for VF test (i.e. Alegret et al. 2012; Olabarrieta-Landa et al. 2015, 2017; Dassanayake et al. 2021; Rivera et al. 2021; Macoir and Hudon 2023). According to García-Escobar et al. (2021), in Spain, the lack of normative data for the Spanish-speaking population has been partially overcome by several normative studies (i.e. Nieto et al. 2008; Peña-Casanova et al. 2009; Casals-Coll et al. 2013; Contador et al. 2016; Olabarrieta-Landa et al. 2017; García-Escobar et al. 2021; Lara et al. 2021; Lubrini et al. 2022; García-Herranz et al. 2022; Iñesta, Oltra-Cucarella and Sitges-Macià 2022; López-Higes et al. 2022).

However, although Spanish is the most spoken language in the country, other languages are also spoken, such as Galician, Basque, Catalan, Aranese, Astur-Leonese, and Aragonese. Therefore, the assessment of bilingual individuals in Spain should consider different linguistic profiles, as individuals may possess varying degrees of proficiency in each language.

This paper focuses specifically on Basque and Catalan languages. The distribution of Basque speakers varies across regions and linguistic zones (Autonomous Community of the Basque Country, Chartered Community of Navarre, and the French Basque Country), and is further influenced by historical and political factors that impact both proficiency and literacy levels. During Franco's dictatorship, the use and teaching of Basque were prohibited, leading to generational gaps in literacy. In contrast, institutional support in recent decades, such as the implementation of bilingual education and the use of *Euskara Batua* (standardized Basque), has promoted language revitalization. Similarly, proficiency and literacy in Catalan also show considerable regional variation. The suppression of Catalan during the Franco regime resulted in gaps in intergenerational transmission and formal literacy, although the linguistic normalization process that began in the late 1970s has significantly promoted its use in education, administration, and media.

Olabarrieta-Landa et al. (2016) found that, among the main problems with neuropsychological instruments detected by neuropsychologists from Spain, the lack of normative data for the country (49%), adaptation to the culture (18%), and translation to their language (17%) arose. Indeed, today normative data for Basque speakers are nonexistent and as a result, this is highly needed. For Catalan speakers, Alegret et al. (2012) obtained normative data for P letter and animals' category for individuals older than 49 years old, so there is a lack of norms for younger individuals. Moreover, there is a lack of verb fluency test norms for this population. Thus, the objective of the present study is to generate normative data for letter, semantic/category, and action/verb VF tests for Basque and Catalan speakers from Spain.

Materials and methods

Participants

The total sample consisted of 275 cognitively healthy adults, 141 Basque speakers from the Autonomous Community of the Basque Country, and the Chartered Community of Navarre, and 134 Catalan speakers from Catalonia, Spain. Inclusion criteria were (a) individuals aged between 18 and 85 years, (b) completion of at least elementary school, (c) a Mini-Mental State Examination (MMSE; Lobo et al. 2002) score ≥ 27 , and (d) a Patient Health Questionnaire-9 (PHQ-9; Kroenke, Spitzer and Williams 2001) score ≤ 4 . Participants were excluded if a diagnosis of history of neurological or psychiatric disorders was reported. Table 1 shows the sociodemographic and linguistic characteristics of the sample.

The desired margin of estimation error (τ) for sample size calculation was established using the formula for optimal sample size in the regression approach (Innocenti et al. 2021). In this approach, covariates/predictors were considered, where the case of the confidence interval of 95%, number of predictors in the final model ($k = 4$), obtaining margin of estimation error (τ) levels of 0.3–0.4.

Table 1. Sociodemographic and linguistic characteristics of the Basque and Catalan samples.

Variable		Basque	Catalan
		Mean (SD)/% (n)	Mean (SD)/% (n)
Age		45.34 (18.90)	49.96 (19.22)
Education		13.73 (3.60)	13.54 (4.14)
Gender	Female	60.3% (85)	63.4% (85)
	Male	39.7% (56)	36.6% (49)
Mother tongue	Spanish	29.1% (41)	35.8% (48)
	Basque	62.4% (88)	53.7% (72)
	Both languages ^a	8.5% (12)	10.4% (14)
Competence	Speaking	8.74 (1.09)	9.66 (0.74)
	Comprehension	9.15 (0.96)	9.84 (0.52)
	Writing	8.26 (1.51)	9.53 (0.92)
	Reading	8.77 (1.34)	9.67 (0.86)
Competence	Speaking	8.73 (1.02)	9.80 (0.54)
	Comprehension	9.06 (0.96)	9.58 (0.78)
	Writing	7.57 (2.27)	8.17 (2.08)
	Reading	8.04 (1.96)	9.34 (1.09)

^aPeople who grow up with both languages (Basque/Catalan and Spanish).

Instruments

Letter VF test

Participants were prompted to generate, within 60 seconds, words that begin with a certain letter (Benton and Hamsher 1989), for this study F, A, S, M, R, P, E, and B. The letters F, A, and S correspond to the original set of letters and the most widely used combination (Olabarrieta-Landa et al. 2017). The choice of the other letters was done considering the specific features of each language. Letters M, R, and P have been proposed for Spanish-speaking individuals (Artiola et al. 1999; Peña-Casanova et al. 2009). The set A, E, and B has been proposed for Basque-speaking people, whereas M, P, and F for Catalan speakers (Olabarrieta-Landa et al. 2019). The letters were selected based on their frequency in each language and were empirically cross-validated through preliminary word production data (Artiola et al. 1999; Peña-Casanova et al. 2009; Olabarrieta-Landa et al. 2019).

Semantic or category VF test

Participants were asked to generate, within 60 seconds, words that belong to a specific semantic category (Benton and Hamsher 1989); for this study animals, fruits, and occupations.

Action or verb fluency test

Examinees were instructed to generate, within 60 seconds, as many verbs in infinitive as s/he can (Piatt et al. 1999).

The specific instruction and scoring guidelines given to participants for each language and task can be found in Appendices 1–2. For all VF forms, the total number of correct words was obtained.

Procedure

Two groups of Basque and Catalan researchers and psychologists were responsible for the recruitment and assessment of participants. Examiners were Basque/Spanish- or Catalan/Spanish-fluent bilinguals, capable of administering and scoring the test in both languages. Participants were recruited from the general population through flyers at neighborhood, universities, and stores, and by general word of mouth. All participants completed a sociodemographic questionnaire and the screening tests (PHQ-9 and MMSE). Those participants who met inclusion and exclusion criteria completed the three VF tests for each language: Basque speakers in Basque and Spanish, and Catalan speakers in Catalan and Spanish, with counterbalanced tests order across participants. Each language was administered in a separate trial, with instructions provided in the corresponding language (see Appendices 1–2). The protocol was administered in a single assessment. Before testing administration, each participant completed and signed an informed consent. Participants did not receive any reward for participating. This study was approved by the ethics committee of University of Deusto.

Statistical analysis

Since each participant contributed multiple observations across different task types (e.g., Letter F, Letter B), the effects of demographic variables on VF scores were evaluated using Generalized Linear Mixed Models (GLMMs) with a Poisson distribution. Given that the outcome variable Y_{ij} corresponds to the number of correct words produced by individual i in task j , a Poisson distribution was assumed $Y_{ij} \sim \text{Poisson}(\mu_{ij})$, with

$$\log(\mu_{ij}) = \beta_0 + b_{0i} + \beta_1 X_{1ij} + \dots + \beta_k X_{kij},$$

where $X_{1ij}, \dots, X_{kij}$ represent the values of the covariates for individual i and task j , $\beta_1, \dots, \beta_k$ are their associated fixed-effect coefficients, and b_{0i} is the subject-specific random intercept.

Three GLMMs (letter, semantic, and verbs) were computed by each group (Basque and Catalan). The full models included the following predictors: language (Spanish, coded as 0, vs. regional language, coded as 1), age, age², formal education (log transformed), sex, type [letters (for the letter VF) or categories (for the semantic VF)], and interactions between language and type. A second-order polynomial of age was used to model a possible quadratic effect. A Bayesian approach was adopted, wherein the available prior information was combined with the information from the data. This combination is summarized in a posterior probability distribution for each unknown quantity in the model. Prior distributions for each of the unknown parameters ($\beta_0, \beta_1, \dots, \beta_k$) followed a normal distribution centered at 0 and with large variance (i.e. $\sigma^2 = 1.0 \times 10^4$).

The Bayesian inference procedure was performed using Markov Chain Monte Carlo methods through the software JAGS (Just Another Gibbs Sampler) and its R interface *rjags* (Plummer 2025). To ensure convergence, three chains with a 10% of the iterations as burning process were considered. A total of 3,000 samples were kept for each posterior distribution.

Whenever a parameter did not show convergence or its posterior distribution was centered at 0, the model was conducted again without the corresponding predictor, and deviance information criterion (DIC) was considered for comparisons with the full model. If no difference between the parameters of the two models were found, the full model was selected.

A Bayesian approach was used due to its advantages over traditional frequentist methods, especially in how it handles uncertainty and adapts to the data. Unlike frequentist models, which often rely on mathematical assumptions that only hold when the sample size is very large (known as asymptotic assumptions), Bayesian models work directly with the observed data, regardless of sample size (Kruschke 2015). This is particularly important in clinical research, where samples are often small or unbalanced. Bayesian models provide full probability distributions for the expected test scores and for individual percentiles, allowing clinicians to see not just a single estimate, but the range of possible outcomes and how likely each one is. This richer and more transparent information improves clinical interpretation and supports more informed decisions (Jones and Love 2011; Rouder, Morey and Wagenmakers 2016).

Results

Basque sample

Table 2 presents the mean, median, and credible interval (95%) of the posterior distributions for the exponential of the parameters in each model. The main model for letter VF test showed an interaction between language and letter, so that people produced more words in all letters when they performed the task in Spanish than in Basque. The only exception was in letter B, where people achieved to produce more words when tested in Basque, and in letters A and E, where the scores achieved in both languages were similar (see Fig. 1A). Formal education was also related to the letter VF score, so that the number of words produced increased according to education's natural logarithmic transformation. Moreover, age and age² effects arose, so that scores increased

Table 2. Mean, median, and credibility interval (95%) for the posterior distributions of the parameters in the Basque verbal fluency models.

Parameter		Full model				Model 2			
		Mean	Median	HDI low	HDI high	Mean	Median	HDI low	HDI high
Phonologic	Intercept	0.885	0.896	0.421	1.406	0.928	0.924	0.492	1.370
	Language (Basque)	-0.056	-0.056	-0.128	0.009	-0.056	-0.056	-0.123	0.009
	Age	-0.339	-0.342	-1.090	0.414	-0.280	-0.284	-1.026	0.500
	Age ²	-0.856	-0.855	-1.603	-0.071	-0.870	-0.865	-1.606	-0.061
	log(Education)	0.592	0.587	0.413	0.768	0.590	0.592	0.416	0.754
	Sex (Men)	0.027	0.027	-0.062	0.121	-	-	-	-
	Letter B	-0.184	-0.184	-0.258	-0.118	-0.184	-0.183	-0.250	-0.106
	Letter E	-0.046	-0.045	-0.112	0.020	-0.047	-0.048	-0.118	0.020
	Letter F	0.087	0.088	0.024	0.152	0.088	0.087	0.018	0.152
	Letter M	0.193	0.193	0.133	0.257	0.192	0.192	0.132	0.260
	Letter P	0.277	0.277	0.217	0.340	0.277	0.276	0.215	0.337
	Letter R	0.178	0.178	0.112	0.240	0.177	0.176	0.113	0.238
	Letter S	0.157	0.158	0.092	0.221	0.157	0.157	0.087	0.217
	Language * Letter B	0.293	0.293	0.196	0.388	0.293	0.293	0.200	0.391
	Language * Letter E	0.051	0.051	-0.050	0.146	0.052	0.052	-0.046	0.143
	Language * Letter F	-0.868	-0.869	-0.973	-0.763	-0.868	-0.869	-0.974	-0.748
	Language * Letter M	-0.277	-0.278	-0.369	-0.183	-0.277	-0.276	-0.369	-0.181
	Language * Letter P	-0.443	-0.443	-0.536	-0.350	-0.443	-0.443	-0.542	-0.352
	Language * Letter R	-2.332	-2.332	-2.495	-2.172	-2.333	-2.331	-2.502	-2.182
	Language * Letter S	-0.526	-0.525	-0.627	-0.432	-0.527	-0.528	-0.631	-0.431

Table 2. Continued

Parameter		Full model				Model 2			
		Mean	Median	HDI low	HDI high	Mean	Median	HDI low	HDI high
Semantic	Intercept	2.157	2.155	1.873	2.452	2.154	2.152	1.867	2.475
	Language (Basque)	-0.185	-0.185	-0.236	-0.126	-0.191	-0.191	-0.226	-0.155
	Age	-0.864	-0.862	-1.315	-0.365	-0.881	-0.877	-1.354	-0.408
	Age ²	-1.161	-1.160	-1.634	-0.676	-1.172	-1.171	-1.668	-0.686
	log(Education)	0.312	0.311	0.210	0.420	0.314	0.315	0.200	0.425
	Sex (Men)	0.017	0.016	-0.046	0.073	0.017	0.017	-0.044	0.071
	Fruits	-0.425	-0.426	-0.482	-0.367	-0.421	-0.421	-0.463	-0.379
	Occupations	-0.300	-0.300	-0.357	-0.245	-0.311	-0.311	-0.350	-0.269
	Language * Fruits	0.006	0.006	-0.083	0.092	-	-	-	-
	Language * Occupations	-0.024	-0.024	-0.105	0.062	-	-	-	-
Action	Intercept	1.151	1.154	0.571	1.710	2.210	2.211	1.928	2.499
	Language (Basque)	-0.151	-0.151	-0.205	-0.093	-0.190	-0.190	-0.225	-0.156
	Age	-1.542	-1.533	-2.496	-0.595	-0.845	-0.850	-1.327	-0.392
	Age ²	-0.210	-0.217	-1.204	0.691	-1.188	-1.188	-1.680	-0.695
	log(Education)	0.706	0.705	0.490	0.906	0.302	0.302	0.189	0.408
	Sex (Men)	-0.040	-0.040	-0.157	0.080	-	-	-	-

HDI = highest density intervals. DIC phonologic = -0.26; DIC semantic = 3.84; DIC action = 0.21. Bold values indicate variables retained in the final model after DIC comparison.

until 40 years old, then started to decrease slowly (see Fig. 1B). Finally, sex was found to be related to letter VF score, so that men scored higher compared to women.

The model 2 for semantic VF test showed that formal education and sex were related to the performance on the test, so that people increased the production of words according to education's natural logarithmic transformation, and men produced more words than women. Moreover, age and age² effects arose, so that scores increased until 40 years old, then started to decrease slowly. Finally, language and category were found to be related to semantic VF performance, so that people produced more words when they were tested in Spanish, and said more animals compared to fruits or occupations.

Regarding verb VF test, the main model showed that education and sex were related to the performance on this test, so that people with high education produce more words (see Fig. 1D), and woman produced more words than men. Moreover, there was a main effect for age and age², so that the number of words decreased slowly while people get older (see Fig. 1C). Finally, there was a main effect of language, so that people produced more words when they were tested in Spanish than in Basque.

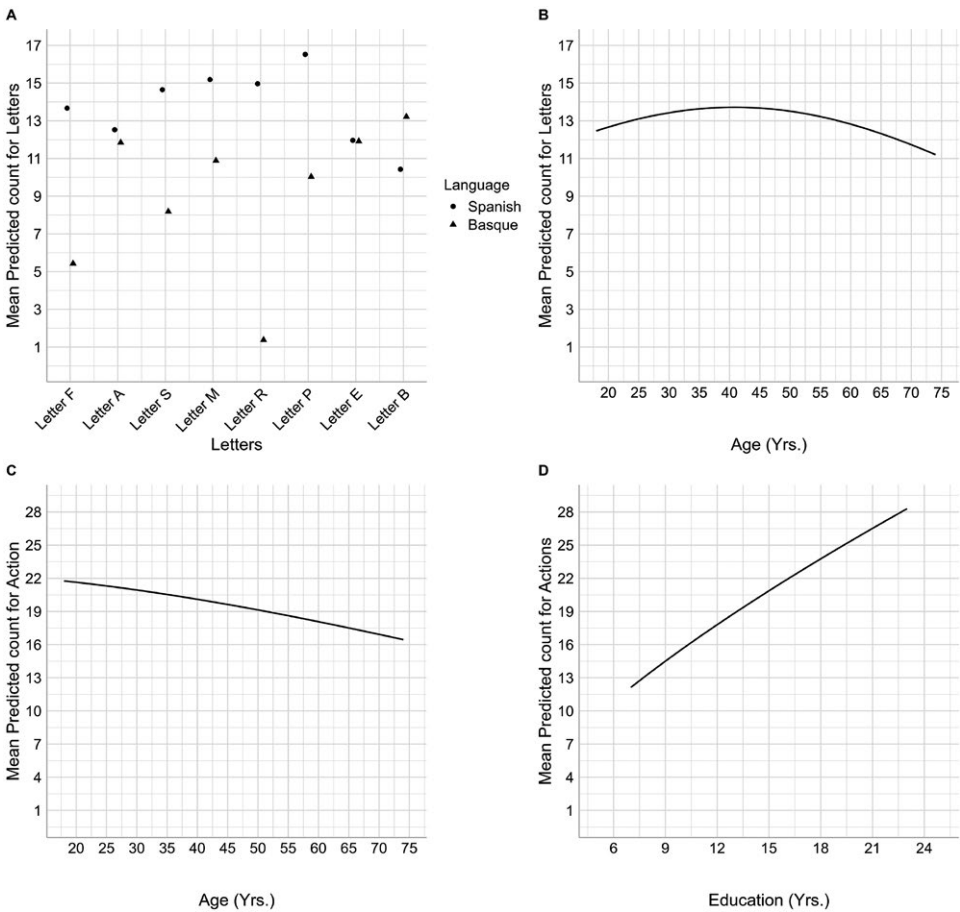


Figure 1. Predicted mean count as a function of demographic variable included in the VF test models in the Basque sample.

Catalan sample

Table 3 presents the mean, median, and credibility interval (95%) of the posterior distributions for the exponential for the parameters in each model. The main model for letter VF test showed an interaction between language and letter, so that people produced more words when they performed the task in Spanish than in Catalan. The only exception was in letters F, B, and E where the scores were similar in both languages (see Fig. 2A). Formal education was related to the letter VF scores, so that the number of words produced increased slowly and constantly according to years of formal education (see Fig. 2C). Moreover, there was a main effect for age and age², so that the number of words increased until 45 years old, then started to decrease (see Fig. 2B). Finally, sex was found to be related to letter VF total score, so that men produced more words than women.

The model 2 for semantic VF test showed that number of years of education and sex were related to the test total scores, so that people produced more words according to the number of years of formal education's natural logarithmic transformation, and women produced more words than men. Moreover, there was a main effect for age and age², so that total number of words increased until 45 years old, then started to decrease. Finally, language and category were

found to be related to semantic VF scores, so that people scored higher when they were tested in Spanish, and achieved higher scores in animals' category compared to fruits and occupations.

The main model for verb VF test showed that number of years of formal education and sex were related to the total scores, so that people with higher number of years of education produced more words (see Fig. 2E), and woman produced more words than men. Moreover, there was a mean effect for age and age², so that scores increased until 45 years old and then decreased (see Fig. 2D). Finally, language was related to total scores, so that people produced more words when they were tested in Spanish than in Catalan.

Table 3. Mean, median, and credibility interval (95%) for the posterior distributions of the parameters in the Catalan verbal fluency models.

Parameter		Main model				Model 2			
		Median	HDI low	Median	HDI low	Median	HDI low	Median	HDI low
Phonologic	Intercept	2.023	2.013	1.655	2.444	2.015	2.014	1.615	2.384
	Language (Catalan)	-0.203	-0.203	-0.272	-0.133	-0.205	-0.204	-0.277	-0.135
	Age	-0.581	-0.580	-1.407	0.150	-0.570	-0.572	-1.359	0.196
	Age ²	-2.694	-2.692	-3.443	-1.975	-2.676	-2.680	-3.430	-1.938
	log(Education)	0.145	0.149	-0.013	0.287	0.180	0.183	0.039	0.330
	Sex (Men)	0.058	0.058	-0.024	0.155				
	Letter B	-0.177	-0.177	-0.247	-0.108	-0.178	-0.179	-0.248	-0.104
	Letter E	-0.132	-0.132	-0.202	-0.061	-0.134	-0.135	-0.201	-0.062
	Letter F	-0.048	-0.048	-0.116	0.021	-0.049	-0.049	-0.117	0.018
	Letter M	0.098	0.097	0.033	0.162	0.096	0.095	0.025	0.158
	Letter P	0.236	0.236	0.173	0.299	0.277	0.276	0.215	0.337
	Letter R	0.102	0.102	0.040	0.171	0.101	0.100	0.037	0.166
	Letter S	0.100	0.101	0.035	0.165	0.099	0.099	0.031	0.166
	Language * Letter B	0.112	0.112	0.009	0.216	0.112	0.113	0.010	0.219
	Language * Letter E	0.127	0.127	0.028	0.231	0.128	0.127	0.025	0.227
	Language * Letter F	0.245	0.244	0.151	0.346	0.247	0.247	0.148	0.346
	Language * Letter M	0.093	0.093	-0.005	0.188	0.096	0.096	-0.006	0.190
	Language * Letter P	0.072	0.071	-0.028	0.160	0.071	0.071	-0.026	0.164
	Language * Letter R	0.060	0.061	-0.039	0.155	0.062	0.061	-0.031	0.164
	Language * Letter S	0.019	0.019	-0.079	0.120	0.019	0.018	-0.078	0.120

Table 3. Continued

Parameter		Main model				Model 2			
		Median	HDI low	Median	HDI low	Median	HDI low	Median	HDI low
Semantic	Intercept	2.708	2.708	2.418	2.982	2.693	2.689	2.431	2.986
	Language (Catalan)	-0.057	-0.057	-0.110	-0.002	-0.043	-0.043	-0.076	-0.007
	Age	-0.476	-0.473	-0.942	0.024	-0.472	-0.475	-1.023	-0.007
	Age ²	-1.434	-1.433	-1.914	-0.956	-1.443	-1.439	-1.958	-0.988
	log(Education)	0.148	0.148	0.046	0.254	0.150	0.153	0.043	0.249
	Sex (Men)	-0.104	-0.104	-0.168	-0.046	-0.103	-0.103	-0.166	-0.039
	Fruits	-0.353	-0.353	-0.410	-0.291	-0.342	-0.342	-0.385	-0.298
	Occupations	-0.272	-0.272	-0.330	-0.212	-0.259	-0.259	-0.301	-0.217
	Language * Fruits	0.024	0.024	-0.061	0.108	-	-	-	-
	Language * Occupations	0.024	0.024	-0.061	0.106	-	-	-	-
Action	Intercept	2.069	2.068	1.617	2.544	1.948	1.956	1.493	2.365
	Language (Catalan)	-0.116	-0.116	-0.175	-0.057	-0.115	-0.116	-0.172	-0.058
	Age	-1.006	-0.996	-1.896	-0.207	-1.007	-1.009	-1.851	-0.137
	Age ²	-2.122	-2.128	-2.962	-1.326	-2.224	-2.213	-3.117	-1.451
	log(Education)	0.357	0.357	0.178	0.519	0.360	0.358	0.192	0.528
	Sex (Men)	-0.083	-0.083	-0.181	0.020	-	-	-	-

HDI = highest density intervals. DIC phonologic = -2.55; DIC semantic = 4.76; DIC action = 1.59. Bold values indicate variables retained in the final model after DIC comparison.

Normative procedure

Norms (e.g. a percentile score) for the VF total scores were established using a four-step procedure: (1) The expected test score ($\hat{\mu}_i$) was computed based on the parameter estimates from the established regression model ($\hat{\beta}_p$), using:

$$\hat{\mu}_i = e^{\hat{\beta}_0} \cdot e^{\hat{\beta}_1 X_i^{(1)}} \cdot \dots \cdot e^{\hat{\beta}_p X_i^{(p)}}$$

Bear in mind that demographic variables were multiplied by 3,000 samples of parameters. (2) The probability based on the Poisson Cumulative Distribution Function was estimated for each of the 3,000 sample parameters, (3) a mean probability was calculated, and (4) this probability was multiplied by 100 to interpret it as a percentile.

User-friendly normative data

To enhance the clarity and clinical applicability of our normative data, we include below a step-by-step example of how to calculate the percentile rank for a given patient. Consider a Basque-speaking woman, aged 60, with 12 years of formal education, who obtained a raw score of 15 on the animal fluency task administered in Basque. Because age was modeled using orthogonal

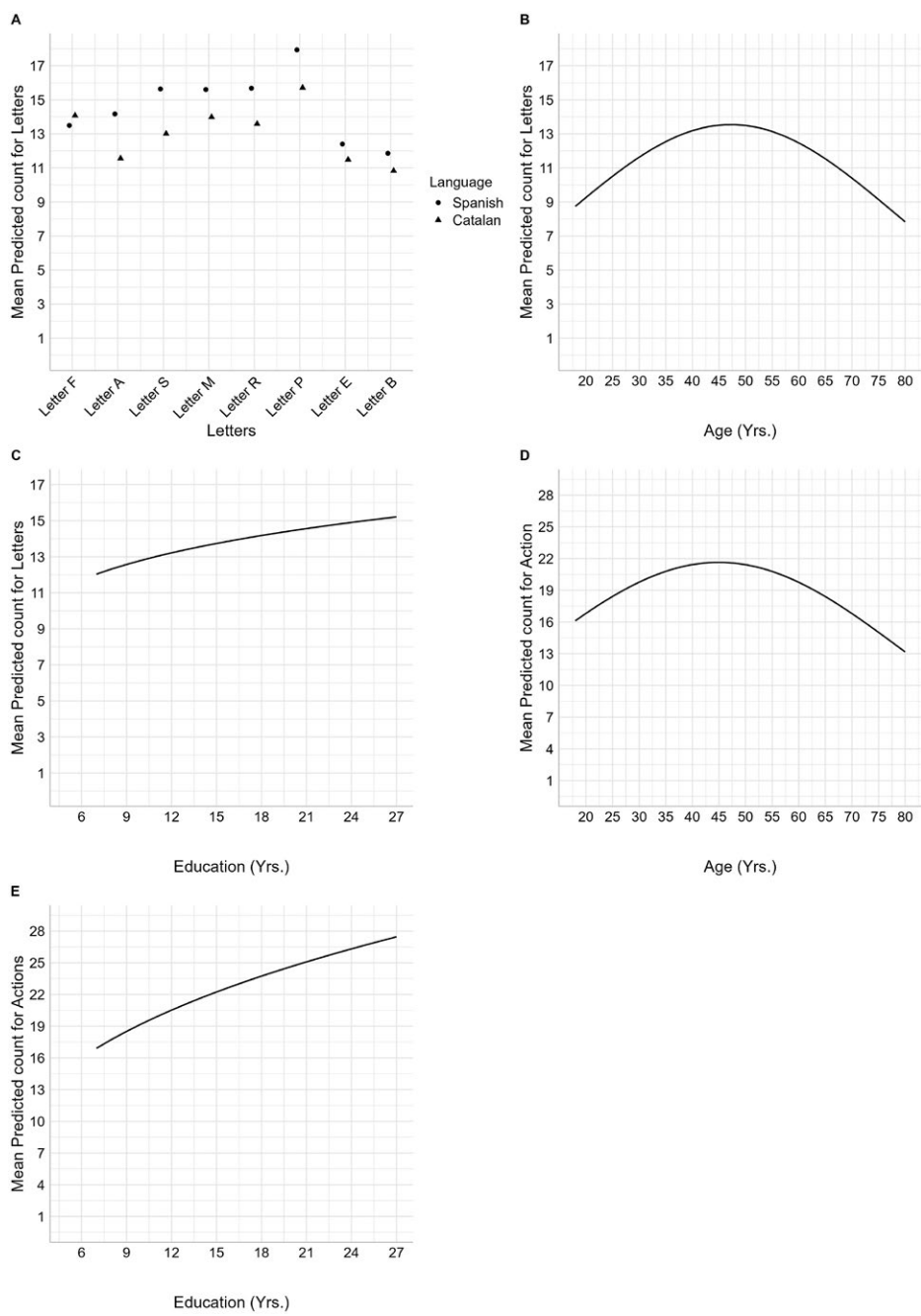


Figure 2. Predicted mean count as a function of demographic variable included in the VF test models in the Catalan sample.

polynomial terms of degree two, we first transform the value of age = 60 using the coefficients derived from the original sample. The corresponding orthogonal components are $Age_1 = 0.0463$ and $Age_2 = -0.0202$. Education was modeled as the natural logarithm of years of schooling, resulting in $\log(\text{Education}) = \log(12) = 2.4849$. Sex was coded as 0 = women, while language as

0 = Basque. The Poisson regression equation used to compute the expected value (λ) for the count distribution is: $\lambda = \exp(\beta_0 + \beta_1 \cdot \text{Language}_i + \beta_2 \cdot \text{Age}_1 + \beta_3 \cdot \text{Age}_2 + \beta_4 \cdot \log(\text{Education}_i) + \beta_5 \cdot \text{Sex}_i)$. For this case, the mean posterior distributions of the parameters were as follows: $\beta_0 = 2.154$, $\beta_1 = -0.191$, $\beta_2 = -0.881$, $\beta_3 = -1.172$, $\beta_4 = 0.314$, and $\beta_5 = 0.017$ (see Table 2). Substituting the patient's values into the equation:

$$\begin{aligned}\lambda &= \exp(2.154 + (-0.191 \cdot 0) + (-0.881 \cdot 0.0320) + (-1.172 \cdot -0.0473) + (0.314 \cdot 2.4849) + (0.017 \cdot 0)) \\ &= \exp(2.154 - 0.0282 + 0.0554 + 0.7802) \\ &= \exp(2.917204) = 18.49.\end{aligned}$$

This λ represents the expected mean of the Poisson distribution for the patient's profile. To compute the probability of obtaining a score of 15 (animals) or lower under this distribution, we use the cumulative function of the Poisson distribution: $P(X \leq 15 | \lambda = 18.49) = 0.2498$. Therefore, the patient's score corresponds to 24.9th percentile. Given the complexity of this calculation and its susceptibility to human error, we have developed an interactive web-based calculator using the Shiny framework (<https://www.rstudio.com/products/shiny/>). This tool enables clinicians to input patient characteristics and test scores and obtain the corresponding percentiles with ease. Since the calculator integrates the full posterior distributions of all model parameters, slight differences may arise between the manually computed percentiles (24.9th in this example) and those produced by the calculator (e.g. 25.3rd percentile). These differences reflect the inherent uncertainty captured by the Bayesian estimation process. The calculator is freely available at https://diegorivera.shinyapps.io/calculator_basq_cat/.

Discussion

This study examined the effects of demographic and language-related variables on VF performance across Basque and Catalan speakers. In both language groups, test performance was generally higher when participants were assessed in Spanish, although some exceptions were observed depending on specific letters or semantic categories. Education consistently showed a positive effect on performance, particularly when modeled as a logarithmic function, while age followed a nonlinear pattern.

Task-specific effects also emerged. For letter fluency, letter-based differences interacted with testing language, suggesting that certain letters may elicit more words depending on the language context. In semantic fluency, participants produced more words in the "animals" category compared to "fruits" or "occupations," regardless of testing language.

According to the literature, currently eight studies (Benito-Cuadrado et al. 2002; Peña-Casanova et al. 2009; Casals-Coll et al. 2013; Contador et al. 2016; Lubrini et al. 2022; García-Herranz et al. 2022; Iñesta, Oltra-Cucarella and Sitges-Macià 2022; López-Higes et al. 2022) provide norms for adults Spanish speakers from Spain, mostly for "F, A, and S", and "P, M, and R" letter combinations, and animals and fruits categories. Thus, as García-Escobar et al. (2021) said, the lack of norms has been partially overcome during the last years. Partially because, these studies have focused mainly on older adults (>50 years old), and all of them provide norms for Spanish native speakers, except Peña-Casanova et al. (2009), Benito-Cuadrado et al. (2002), and Casals-Coll et al. (2013), who recruited Spanish participants from bilingual regions and may have some degree of bilingualism (not specified in the studies).

Today, there is a lack of demographically adjusted neuropsychological test norms for Basque and Catalan speakers. Indeed, and specifically for VF tests, normative data for Basque speakers are nonexistent, and norms for Catalan speakers are scarce, since only Alegret et al. (2012) obtained normative data for the P letter and animals category. Therefore, to our knowledge, this is the first study that provides normative data for three VF tests for these two bilingual populations. Moreover, there is a vast literature that shows how bilingualism is an important factor to consider when generating norms for verbal tasks, particularly VF tests, as bilinguals suffer from between-language interference that impacts their lexical access, and thus, word production (Sandoval et al. 2010). Therefore, it becomes evident that norms

generated for Spanish native speakers may not be appropriate to use on Basque and Catalan bilingual populations.

The results of the study suggest that sociodemographic variables, primarily age (curvilinear) and formal education (natural logarithm), were associated with all VF test scores, so the number of words increased slowly with an increase in years of education and until around 40–45 years old, then decreased later with age. The only exception was action/verb VF in Basque, where the number of words decreased while people get older. These results are in line with previous normative studies in phonological/letter (i.e. [Marquine et al. 2021](#); [Rivera et al. 2021](#); [García-Herranz et al. 2022](#); [Iñesta, Oltra-Cucarella and Sitges-Maciá 2022](#); [Lubrin et al. 2022](#); [Martino et al. 2022](#); [Zegarra-Valdivia, Chino and Paredes Manrique 2022](#)), semantic/category (i.e.; [Marquine et al. 2021](#); [García-Herranz et al. 2022](#); [Iñesta, Oltra-Cucarella and Sitges-Maciá 2022](#); [Jebahi, Abou Jaoude and Ellis 2022](#); [Lubrin et al. 2022](#); [Martino et al. 2022](#); [Zegarra-Valdivia, Chino and Paredes Manrique 2022](#)), and action/verb ([Piatt et al. 2004](#); [Woods et al. 2005](#); [Nogueira, Reis and Vieira 2016](#); [Alegret et al. 2018](#); [Martino et al. 2022](#); [Macoir and Hudon 2023](#)) VF. Regarding sex, discrepancies can be found in the literature regarding its influence, but in this study, it was only associated with semantic/category (Catalan sample) and action/verb fluency test (Basque and Catalan speakers).

Both bilingual participants scored higher when tested in Spanish than in Basque or Catalan, even though for up to 50% of the participants Basque and Catalan represented their mother tongue, and all of them had very good speaking and comprehension competences in both languages, with no significant differences. Similar results have been reported in studies with bilinguals' participants. For example, [de Picciotto and Friedland \(2001\)](#) reported that 20% of their sample performed better in their L2 during semantic/category fluency, even though their participants spoke their mother tongue $\geq 70\%$ of the time. Also, [Jordan \(2014\)](#) found that Lakota-English bilinguals with high Lakota language competence scored lower when participants were tested in Lakota compared to English during letter and semantic/category VF. According to her, minority groups have reduced exposure to their native language compared to surrounding dominant languages, and this may have some impact on VF performance. Basque- and Catalan-speaking regions are smaller compared to Spanish, so that reduced opportunities exist to interact in both outside friends and family environments ([Olabarrieta-Landa et al. 2019](#)).

Moreover, patterns of language use are context-specific and shaped by availability. In the Basque-speaking group, participants tended to use Spanish more frequently for media consumption (e.g. watching television, online reading) or reading for pleasure, likely due to lower availability of content in Basque. In contrast, Catalan speakers reported more use of Catalan for watching television and reading in professional or academic contexts, but tended to use more Spanish when browsing the internet.

It is also important to bear in mind that some of these participants were not schooled in their local language because during Franco's dictatorship, the use and teaching of languages other than Spanish were forbidden. Languages differ in their linguistic features (i.e. grammar, phonology, pragmatics, lexicon, and reading system), which may affect language test performance ([Marquine et al. 2021](#)). Schooling allows literacy in a particular language, and this opens the opportunity to acquire and improve orthography and vocabulary in that language, both important variables in the VF tests ([Olabarrieta-Landa et al. 2019](#)).

Also, sociolinguistic and psycholinguistics aspects may have impact on participants VF test performance during Basque or Catalan evaluations. Indeed, [Jebahi, Abou Jaoude and Ellis \(2022\)](#) said that language interacts with culture and modulate semantic processing, organization, and categorization, such as lexically driven semantic associations. [Olabarrieta-Landa et al. \(2019\)](#) found that Basque participants, but not Catalans, tended to avoid words that were similar to Spanish when tested in the Basque language as they judged these words as "unacceptable." For example, "baso" ("glass" in English, "vaso" in Spanish) may be seen by some speakers as inadequate, and therefore, they would try to find another word, generally from the Standardized Basque language ("edalontzi"). As such, participants were, unwittingly, making the task even

more complex and difficult, losing time searching for “appropriates” words. This kind of additional cognitive demand has been also reported before in other languages. For example, [Muayqil et al. \(2021\)](#) reported an anecdotal experience of how their Arabic-speaking participants suffered for higher cognitive demands during letter VF as adjectives could be used for people's names (not allowed during production), so participants had to filter out these adjectives mentally during production.

Another interesting finding was found for letter such that language-by-letter interactions were found in Basque but also in Catalan samples. As pointed out before, higher scores were, in general, achieved in Spanish than in Basque or Catalan; moreover, participants performed better in M, R, and P letters when tested in Spanish, while in Basque, the highest scores were obtained with A, E, and B letters, and in Catalan with P, F, and M. It is well known that the frequency of letters varies along languages, so that the most suitable combination of letters should be chosen of each language. These results are in line with [Olabarrieta-Landa et al.'s \(2019\)](#) suggestion of using A, E, and B combinations for Basque speakers, and M, P, and F for Catalan speakers.

The information above highlights how neuropsychological verbal measures normed on a specific group may not be valid for assessing individuals from other cultural and linguistically diverse backgrounds ([Jebahi, Abou Jaoude and Ellis 2022](#)). To further exemplify the impact of having norms for a given population, an example is provided. Let's imagine a 60-year-old Spanish-Catalan bilingual women with 12 years of formal education who generated 15 animal words during a semantic task when tested in Spanish. Using normative data for monolingual Spanish native population (obtained from [Lubrin et al. 2022](#)), this woman would obtain a percentile of 75. However, with present norms, this woman would obtain a percentile of 25. Thus, using norms from other population has severe implications since the women would go from a “High average score” to an “Average score” performance status ([Guilmette et al. 2020](#)).

Limitations

The results of this study should be interpreted in light of the following limitations. First, these norms are applicable for Basque bilinguals from Spaniard region; French regional bilingual participants were not included. The same applies with Catalan bilingual participants, who were recruited only from Catalonia, excluding other regions where Catalan is spoken (i.e. France, Balearic Islands). Second, exposure to and use of each of the languages were not equal among participants. Some participants came from predominantly Spanish-speaking areas, limiting their opportunity to interact in Basque or Catalan compared to participants living in mostly Basque- or Catalan-speaking regions. These regional differences in language exposure and, importantly, in schooling models and literacy, may have influenced participants' performance on VF tasks, particularly in lexical retrieval and vocabulary range. Importantly, in a previous study with this same sample ([Olabarrieta-Landa et al. 2019](#)), no significant differences were found in VF performance between participants who had Basque or Catalan as their native language and those who acquired it later, after controlling for linguistic competence. This suggests that, at least in the context of our sample, language dominance or order of acquisition did not have a major impact on performance, and that participants achieved comparable scores regardless of whether Basque or Catalan was their first or second language.

Lastly, although the total sample size was large, subsample size, according to different language groups, may be limited. However, the present study used a Bayesian approach that uses precision as parameter, analogous to accuracy in parameter estimation at frequentist approach. But Bayesian highest density interval does not suffer from the instability as of frequentist confidence intervals when testing or sampling intentions change ([Kruschke 2015](#)).

Implications

The results of the present study have several implications. First, it will be expected that clinical neuropsychologists from these two regions will start using these norms when they have to

evaluate VF on bilingual clients. Second, researchers should work toward the creation of norms for another neuropsychological test on these two populations. This will help to improve the diagnosis and treatment of individuals with different neurological conditions on these two regions. Also, this study fills a substantial gap regarding verb fluency, as [Macoir and Hudon \(2023\)](#) stated, few norms exist for this VF form, specifically for English- (in the USA; [Piatt et al. 2004](#); [Woods et al. 2005](#)), Portuguese- ([Nogueria et al. 2016](#)), Danish- ([Vogel, Stokholm and Jørgensen 2020](#)), Swedish- ([Tallberg et al. 2008](#)), and Spanish-speaking populations ([Casals-Coll et al. 2013](#)). Finally, Bayesian approach has been used. Its advantage relays on a better consideration of the uncertainty surrounding the data. Having a posterior distribution as results allow to obtain not only the full posterior distribution for the expected number of words, but also the full distribution for the percentile of a given individual. This posterior distribution can be summarized with its mean or median, but it also considered all its uncertainty for a better understanding of the individual situation.

Conclusion

This study addresses an important gap of the literature since it provides VF norms for Basque and Catalan bilingual speakers. These norms fit an important need for clinical neuropsychologists in Spain, so that now they can evaluate their clients in their preferred or both languages. As so, we hope it may further support to develop culturally sensitive evaluations.

Notes on Contributors

Dr. Laiene Olabarrieta-Landa is a professor in the Department of Health Sciences at Universidad Pública de Navarra, Spain. Her main research interests focus on bilingualism, neuropsychological assessment, and acquired brain injury. In this study, she contributed to conceptualization, data curation, investigation, methodology, project administration, supervision, validation, visualization, writing – original draft, and writing – review & editing.

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Dr. Montserrat Alegret is currently the Coordinator of the Clinical Neuropsychology Unit at Fundació ACE. She has extensive experience in the diagnosis, clinical management, neuropsychological assessment, cognitive stimulation, and research of various neurodegenerative diseases. She has been involved in several projects focusing on the early stages of neurodegenerative diseases, including Parkinson's, Huntington's, and Alzheimer's diseases. Additionally, she has two patents (FACEmemory and NBACE). In this study, she contributed to investigation, methodology, resources, validation, and writing – review & editing.

Alba Pérez is a neuropsychologist at Fundació ACE. Her research interests focus on neuropsychology, and neurodegenerative diseases, especially Alzheimer's disease. In this study, she contributed to investigation, resources, and writing – review & editing.

Dr. Anabel Forte is a full professor at University of Valencia. Her research focuses on the development of statistical techniques that allow us to understand the best way to mathematically model a process of interest. She is currently the president of the Spanish Society of Biostatistics. In this study, she contributed to methodology, supervision, visualization, and writing – review & editing.

Dr. Juan Carlos Arango Lasprilla is a IKERBASQUE researcher in Department of Cell Biology and Histology of the University of the Basque Country. His areas of expertise include brain injury, neurorehabilitation, and cross-cultural issues. In this study, he contributed to project administration, supervision, validation, and writing – review & editing.

Conflict of interest

The authors report there are no competing interests to declare.

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Data availability

Data are not publicly available for reasons of data protection.

Appendix 1. Specific instructions given in each language for each verbal fluency task.

Letter verbal fluency

Basque language

“Nik alfabetoko hizki batzuk esango dizkizut. Orduan, hizki horrekin hasten diren ahalik eta hitz gehien esango dizkidazu, ahalik eta arinen; ez dut nahi pertsona, hiri edo herrialdeen izenak erabiltzerik; gainera, handigarriek eta txikigarriek ez dute balio. Galderaren bat? Hasi.”

Time for each letter: 60 seconds

Prompt given to the participant if no response is given within 10 seconds:

Letter F: fama	Letter A: ama	Letter S: sua	Letter M: musika
Letter R: robota	Letter P: pilota	Letter E: eraso	Letter B: besoa

Catalan language

“Li diré unes lletres de l’alfabet. Vull que em digui totes les paraules que comencin per aquella lletra tant ràpid com pugui; jo no vull que utilitzis noms propis de persones, països o ciutats; tampoc utilitzi paraules diminutives ni augmentatives. ¿Alguna pregunta? Comenci.”

Time for each letter: 60 seconds

Prompt given to the participant if no response is given within 10 seconds:

Letter F: fama	Letter A: anell	Letter S: sol	Letter M: mare
Letter R: rei	Letter P: paper	Letter E: elefant	Letter B: bomba

Spanish language

“Yo diré unas letras del alfabeto. Entonces, quiero que me diga todas palabras que inicien con esa letra tan rápido como usted pueda; yo no quiero que use palabras de nombres propios de personas, países o ciudades; además no use palabras diminutivas o aumentativas. ¿Alguna pregunta? Comience.”

Time for each letter: 60 seconds

Prompt given to the participant if no response is given within 10 seconds:

Letter F: fama	Letter A: anillo	Letter S: sol	Letter M: madre
Letter R: rey	Letter P: papel	Letter E: empezar	Letter B: bomba

Semantic or category verbal fluency

Basque language

“Talde edo kategoría bateko izena esango dizut eta talde horri dagokion hitzak esan beharko dizkidazu ahalik eta azkarren. Adibidez, barazkiak esango banizu, orduan espinakak, letxuga, etab. esango zenidake.

Ulertu al duzu? Animalien izenak esadazu. Hasi”

“Ondo, orain fruten izenak esadazu. Hasi”

“Ondo, orain lanbideen izenak esadazu. Hasi”

Time for each category: 60 seconds

Catalan language

“Li diré el nom d’un grup o categoria i vull que em digui totes les paraules que pertanyin a aquell grup tan ràpid como pugui. Per exemple, si li digués vegetals podria dir-me blat, espinagues, enciam, etc. Ho ha entès?. Digui’m noms d’animals. Comenci.”

“Be, ahora digui’m noms de fruites. Comenci”

“Be, ahora digui’m noms de professions i oficis. Comenci”

Time for each category: 60 seconds

Spanish language

“Voy a decirle el nombre de un grupo o categoría y quiero que me diga todas las palabras que pertenezcan a ese grupo tan rápido como pueda. Por ejemplo, si le dijera vegetales podría decirme maíz, espinacas, lechuga, etc. ¿Lo ha entendido? Dígame nombres de animales. Comience.”

“Bien, ahora dígame nombres de frutas. Comience”

“Bien, ahora dígame nombres de profesiones y oficios. Comience”

Time for each category: 60 seconds

Semantic or category verbal fluency

Basque language

“Mesedez, esadazu, ahalik eta azkarren, bururatzten zaizkizun aditz guztiak. Gogoratu ezin dituzula errepi-katu.” Galderaren bat?”

Time for each category: 60 seconds

Prompt given to the participant if no response is given within 10 seconds: abestu.

Catalan language

“Vostè té un minut per dir la major quantitat possible de verbs en infinitiu, és a dir que acabin en ar, er, ir. Recuerdi que no n’ha de repetir cap”

Time for each category: 60 seconds

Prompt given to the participant if no response is given within 10 seconds: cantar.

Spanish language

“Usted dispone de un minuto para mencionar la mayor cantidad posible de verbos o acciones en infinitivo, es decir que terminen en ar, er, ir. Recuerde que no debe repetir ninguno de ellos.”

Time for each category: 60 seconds

Prompt given to the participant if no response is given within 10 seconds: cantar.

Appendix 2. Scoring guidelines for each verbal fluency task.

General rules

- Proper names, intrusions, perseverations, augmentative or diminutive forms, and grammatical variations of gender (e.g. *perro*–*perra*/dog), number (e.g. *perro*–*perros*/dog–dogs),

- person (e.g. *como-comes/eat-eats*), and tense (e.g. *como-comi/eat-ate*) were not accepted. Variations of gender applied only to Catalan and Spanish, as Basque does not mark gender.
- Derivations were accepted as long as they were used to create new words with different meanings (e.g. *pan-panadero-panaderia/bread-baker-bakery*). However, if the addition of suffixes/prefixes did not substantially change the meaning of the word, they were not accepted (e.g. *fácil-fácilmente/easy-easily*).
 - Homonyms, words that are written the same way but have different meanings (e.g. *alce-alce/raising-moose*) or words that are spelled differently but are pronounced the same (e.g. *hierva-hierba/boil-grass*), were accepted as long as the examinee clarifies, during the task or once it was finished, the different meanings or spelling of the words.
 - Compound words were also accepted.
 - Finally, words are accepted if they are commonly used, regardless of whether they appear in the dictionary. This would include words borrowed from other languages (e.g. parking) or jargon words.

Letter verbal fluency

The task should be scored according to spelling and not phoneme/sound.

Semantic or category verbal fluency

Superordinate words (e.g. *pez/fish*) were accepted as long as the examinee did not include words or examples that belong to the same superordinate category (e.g. *merluza/hake, tiburón/shark, sardine/sardine*). Words that mean two sexes of the same animal were considered valid (e.g. *vaca-toro/cow-bull*). Similarly, words that represent the same animal in different developmental stages were considered valid (e.g. *ternera-vaca/calf-cow*). Extinct and magical/mythological animals were also accepted.

Verb or action verbal fluency

Only infinitive verbs were accepted.

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